Pimpri-Chinchwad Education Trust's
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Assignment No 6

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Subject: Design and Analysis of Algorithms
Batch: A3
GitHub Link: https://github.com/SOHAM0007-CODER/DAA_ASSIGNMNET

1. Topic

Select essential earthquake-relief packages for a truck of capacity W kg. Each item has a weight and utility value. Use 0/1 Knapsack to maximize utility without exceeding W: take each package once or leave it; do not split it.

2. Course Objective

- Understand the computational complexity of algorithms.
- Select a suitable algorithm design strategy for a practical problem.

3. Course Outcome

- Apply dynamic programming to an exact single-truck resource-allocation problem.
- Reconstruct selected packages and compare DP with brute force and greedy selection.

Theory

0/1 Knapsack makes two choices for each package: include it or exclude it. The table dp[i][w] stores the highest utility using the first i items and capacity w. Utility is an importance score, not a price. Using the previous row when taking an item prevents selecting it more than once.

If an item is too heavy, leave it. Otherwise, compare the previous answer with the item's utility plus the best answer for the remaining capacity. The simple program handles one truck. Higher utility can encourage critical supplies, but it does not force their inclusion.

Working of the Algorithm

- 1. Read names, integer weights, utilities and one truck capacity.
- 2. Compute the take/leave DP table using the previous item row.
- 3. Backtrack the table and label every package Loaded or Left behind.
- 4. For small inputs, print the full DP table; show truck contents and totals.

Complexity Analysis

The table contains (N+1)(W+1) states, and each state needs constant work. Time is O(N(W+1)), usually written O(NW) for positive W. This is pseudo-polynomial in numeric capacity.

Time Complexity

Best Case: O(N(W+1)).

Average Case: O(N(W+1)).

Worst Case: O(N(W+1)); all table states are filled.

Space Complexity

O(N(W+1)) for the complete table and selected-item reconstruction. Value-only DP can reduce table space to O(W), but this version keeps reconstruction simple.

Algorithm - Steps

- Step 1: Read item count, truck capacity, and each item's name, weight and utility.
- Step 2: Fill the table using the take-or-leave decision.
- Step 3: Backtrack and print selected items, used weight and maximum utility.

Pseudocode

```
dp[0][w] = 0 for every capacity w
for item i from 1 to N:
    for capacity w from 0 to W:
        dp[i][w] = dp[i-1][w]
        if weight[i] <= w:
            take = utility[i] + dp[i-1][w-weight[i]]
            dp[i][w] = max(dp[i][w], take)
backtrack from dp[N][W] to identify selected items
```

Flow of Execution

START -> Read packages and W -> Fill DP table -> Recover selected items -> Print result -> END.

6. Example

Capacity is 15 kg. Blankets (5,30), Food (6,48), BabyFood (3,35), Medicine (4,60), Water (7,28). Select Food, BabyFood and Medicine: weight 13 kg, utility 143, unused capacity 2 kg. All packages are indivisible; the selection table shows take=0 or 1.

Brute force takes O(2^N) time; ratio greedy takes O(N log N) but is not always exact for 0/1 items. With W=50 and items (10,60), (20,100), (30,120), greedy gives 160 while DP gives 220. For several trucks, sequential planning is a heuristic; exact joint planning needs additional capacity dimensions.

Conclusion

The DP finds the maximum utility for one truck and reconstructs whole-package choices. Named supplies, selection decisions and a small DP table make the result easy to explain. Multiple trucks and mandatory priority need additional constraints.

Assignment 6 | SAMPLE INPUT (input.txt)

```
5 15
Blankets 5 30
Food 6 48
BabyFood 3 35
Medicine 4 60
Water 7 28
```

Assignment 6 | FULL PROGRAM OUTPUT

```
ASSIGNMENT 6: DISASTER RELIEF TRUCK - 0/1 KNAPSACK
Enter number of items and truck capacity:
Enter each item: name weight utility

SUPPLY AND SELECTION TABLE
Name | Weight kg | Utility | Divisibility | Take (0/1) | Decision
Blankets | 5 | 30 | Whole package only | 0 | Left behind
Food | 6 | 48 | Whole package only | 1 | Loaded
BabyFood | 3 | 35 | Whole package only | 1 | Loaded
Medicine | 4 | 60 | Whole package only | 1 | Loaded
Water | 7 | 28 | Whole package only | 0 | Left behind

DP TABLE: maximum utility for each capacity
Item/Capacity | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
No items | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0
Blankets | 0 | 0 | 0 | 0 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30
Food | 0 | 0 | 0 | 0 | 0 | 30 | 48 | 48 | 48 | 48 | 48 | 78 | 78 | 78 | 78 | 78
BabyFood | 0 | 0 | 0 | 35 | 35 | 35 | 48 | 48 | 65 | 83 | 83 | 83 | 83 | 83 | 113 | 113
Medicine | 0 | 0 | 0 | 35 | 60 | 60 | 60 | 95 | 95 | 95 | 108 | 108 | 125 | 143 | 143 | 143
Water | 0 | 0 | 0 | 35 | 60 | 60 | 60 | 95 | 95 | 95 | 108 | 108 | 125 | 143 | 143 | 143

ITEMS IN THE TRUCK
Name | Weight kg | Utility
Food | 6 | 48
BabyFood | 3 | 35
Medicine | 4 | 60
Truck capacity: 15 kg
Total weight: 13 kg
Unused capacity: 2 kg
Maximum utility: 143
Selected packages: 3 of 5
Time and space: O(N*(W+1)) | Each package used at most once.
```